NYC Airbnb Market Segmentation: Identifying Strategic Listing Profiles

Question

NYC has nearly 49,000 Airbnb listings, but treating them as one big group hides very different behaviours underneath. This report asks if listings can be grouped into meaningful segments, and what those segments mean for product, regulation, and growth. The point matters because what works for a casual host does not work for a commercial one. A single platform-wide policy will always under-serve someone.

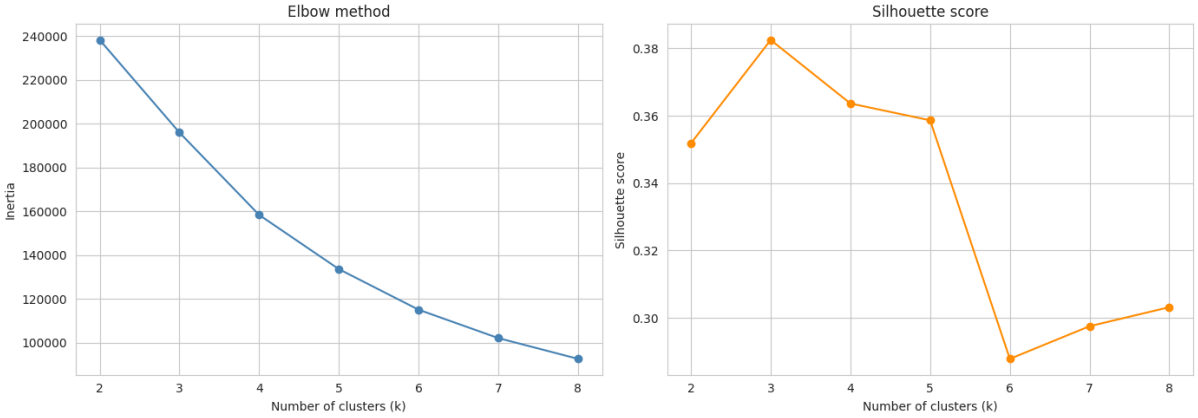
Earlier research on Airbnb in NYC has focused mostly on commercial operators because of their effect on housing supply. Wachsmuth and Weisler (2018) estimated that hosts with multiple listings earn around 28% of the platform's revenue, and split hosts into two groups: commercial and casual. This report goes a step further. Instead of two groups, it uses four groups based on how listings actually behave, and shows that most listings on the platform are barely active — something the simpler two-group view misses.

Method

I used unsupervised k-means clustering (Lloyd, 1982; Pedregosa et al., 2011) and data preparation using pandas (McKinney, 2010) on the AB_NYC_2019 dataset (Dgomonov, 2019), which contains 48,895 listings. K-means groups listings by similarity across multiple features, without using any pre-defined labels. This is useful here because the goal is to discover host types, not to confirm them.

Six features were chosen to describe how a host behaves rather than what kind of property they have: log price, minimum nights, total reviews, reviews per month, annual availability, and total listings per host. Price was log-transformed because it is heavily right-skewed. Missing review rates were treated as zero, and outliers above the 99th percentile were capped. All features were standardised before clustering.

To choose k, I used the elbow method and silhouette score (Figure 1). k=3 gave a slightly higher silhouette (0.382 vs 0.364), but I selected k=4 because it separates a small but important commercial-operator segment that gets merged into a larger group at k=3. The trade-off was deliberate: slightly less statistical separation, but more useful insight.



A chi-square test of independence confirmed that cluster membership is significantly linked to borough ($\chi^2 = 1413.8$, df = 12, $p < 0.001$). The effect size, measured by Cramér's V =

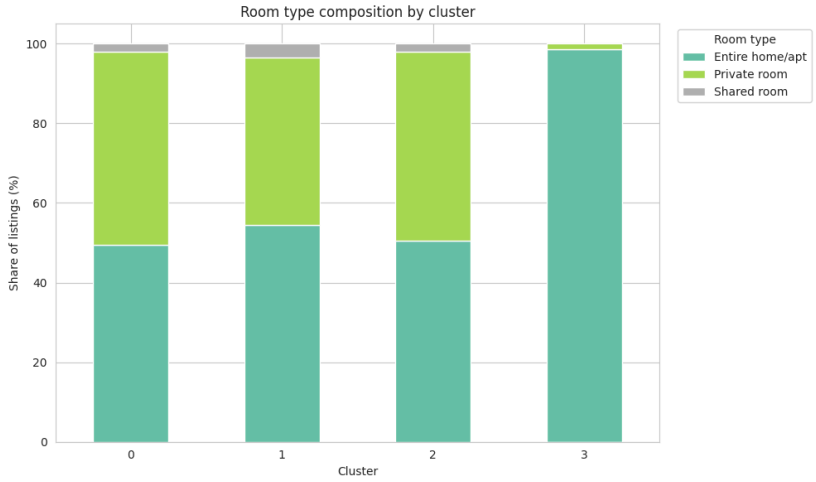
0.098, is weak overall — most of the link comes from Cluster 3 being almost entirely in Manhattan, rather than from clear borough patterns across all segments.

Findings

	listings	median_price	mean_min_nights	mean_reviews	mean_reviews_per_month	mean_availability	mean_host_listings
cluster							
0	7360	99.0	2.5	98.1	3.9	153.7	2.1
1	12163	130.0	14.6	13.9	0.8	290.0	10.2
2	28802	100.0	4.7	8.5	0.5	24.2	1.7
3	559	252.0	21.2	2.3	0.7	281.7	287.6

Four clear segments emerged (Table 1,Figure 2, Figure 3).
Active hosts (Cluster 0 — 7,360 listings, 15%). Median price $99, ~2.5-night minimum stays, 98 total reviews, 3.9 reviews per month. Healthy short-stay rentals — booked often, reviewed well, mostly hosts with one listing spread across all five boroughs. This is the classic Airbnb experience.
Mid-stay multi-listers (Cluster 1 — 12,163 listings, 25%). Median price $130, 14.6-night minimum stays, 290 days available per year, around 10 listings per host, mostly Manhattan. Long minimum stays, almost full-year availability, and multiple listings per host suggest mid-term rental operators — corporate housing, monthly stays, and relocations.
Dormant listings (Cluster 2 — 28,802 listings, 59%). Median $100, only 24 days available per year, 0.5 reviews per month, single-listing hosts. Most listings on the platform are barely active. This is the headline finding. Over half of all listings are essentially inactive but still appear in search results.
Mega-host operators (Cluster 3 — 559 listings, 1%). Median $252, 21-night minimum stays, 282 days available, average of 287 listings per host, 99.6% Manhattan, 98.6% entire homes.

This is property management at large scale, in the borough with the strictest short-term rental laws. Together, this group and the mid-stay multi-listers make up around 26% of listings — close to the 28% commercial revenue share reported by Wachsmuth and Weisler (2018).



Recommendations

On the dormant majority. Most listings are barely active. Airbnb should tell apart listings that are temporarily inactive from those that have been abandoned, using last-review dates and recent availability. Inactive ones can be reactivated through host support; abandoned ones should be hidden from search so they do not get in the way of real listings.

On mega-host operators. This group carries most of the regulatory risk. With 99.6% in Manhattan and 98.6% entire homes, it is exactly the population NYC's Local Law 18 was written to control. It should be treated separately, with stricter checks, required verification, and direct contact with regulators — both to manage risk and to clearly tell apart compliant operators from grey-market ones.

On mid-stay multi-listers. This is a clear product opportunity. The hosts in this group are already running monthly and corporate stays informally. Building proper tools for them — pricing for longer stays, longer-term booking flows, and corporate accounts — would support a use case that already exists on the platform.

On active hosts. This is the group to protect. Better support and search-ranking advantages for highly reviewed listings will keep this core experience strong.

5. Limitations

K-means assumes roughly spherical clusters of similar size. With one cluster holding only 1% of listings, results may be sensitive to feature scaling and how outliers are treated. The dataset is from 2019 and predates NYC's Local Law 18 (2023), which has since reshaped the short-term rental market — the mega-host group in particular is likely smaller today. Geographic coordinates were left out on purpose, to keep the segmentation about behaviour rather than location; a follow-up could add geographic clustering for a richer picture.

Conclusion

Behavioural segmentation shows a marketplace that is far less uniform than the overall numbers suggest. A small commercial group at the top, a large inactive middle, and a healthy active core all need different platform strategies. Treating them as one group risks both regulatory problems and missed opportunities. Treating them as separate segments allows clearer decisions about product, policy, and growth.

References

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